

**MASTER OF COMPUTER
APPLICATIONS (MCA-NEW)**

Term-End Examination

December, 2022

MCS-231 : MOBILE COMPUTING

Time : 3 Hours

Maximum Marks : 100

Note : *Question No. 1 is compulsory and carries 40 marks. Attempt any **three** questions from the rest.*

1. (a) What is Frequency Division Multiplexing (FDM) ? What are its advantages, disadvantages and applications ? 10
- (b) What is GPRS ? Explain its features. Also, list the services offered by it. 10
- (c) What is LTE ? Explain its features. 10
- (d) Write a short note on the Adhoc Networks.

10

2. (a) What is a cellular network ? Explain its various components. 10
(b) Explain mobile computing architecture with the help of a diagram. 10
3. (a) List any *five* applications of mobile computing. 10
(b) Explain J2EE architecture. 10
4. (a) Explain the functions of a mobile operating system. 10
(b) Explain the features of an integrated development platform for mobile apps. 10
5. (a) Explain the features of iOS. 10
(b) Explain the features of XML. 10

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1. (a) What is Time Division Multiplexing (TDM) ? What are the advantages, disadvantages and applications ? 10
 - (b) What is GSM technology ? Explain its advantages and disadvantages. 10
 - (c) What is a 4G network ? Explain its features. 10
 - (d) What is WIMAX ? Explain its features. 10
 2. (a) What is Data Synchronization ? Explain it. 10
 - (b) Write a short note on Mobile Commerce. 10

3. (a) Draw and explain Android architecture. 10
(b) Explain various components of J2ME. 10
4. (a) What is guided transmission ? How does it differ from unguided transmission ? Give example(s). 10
(b) Explain any *two* reasons for weakening of figural strength after transmission. 10
5. (a) What is an antenna ? What is its functionality ? On what factor(s) does its size depend on ? 10
(b) What is CDMA ? How does it differ from GSM ? 10

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1. (a) What is Time Division Multiplexing (TDM)? What are its advantages, disadvantages and applications ? 10
- (b) What is a Cellular Network ? Under what conditions frequency reuse is possible by different cells in the network ? 10
- (c) What are Smart Sensors ? What are the instances where mobile devices interact with their surroundings with the help of sensors ? 10
- (d) Write a short note on Mobile Device Database Management. 10

2. (a) List various steps in the development of a Mobile App. What are the various components of an Enterprise Mobile Application ? 10
(b) Write a short note on XML. 10
3. (a) What is J2ME ? Explain main components of J2ME. 10
(b) Write a short note on Android. 10
4. (a) Explain the features of integrated development platforms that are used for Mobile App development. 10
(b) Explain any *five* features of Mobile Communications. 10
5. (a) Explain the design considerations for Mobile Computing. 10
(b) Write a short note on Smart Appliances. 10

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1. (a) What is Wavelength Division Multiplexing ?
What are its advantages, disadvantages and applications ? 10
- (b) Explain WLAN Wi-Fi IEEE 802.11x networks. 10
- (c) What are Actuators ? Give some examples of Actuators found in Robots. 10

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- (d) Explain the architecture patterns for mobile device data store methods. 10
2. (a) Explain the architecture of a content management application. 10
- (b) What do you mean by XML parsing ? Explain with an example. 10
3. (a) Explain the architecture of J2EE. 10
- (b) Write a short note on SWIFT. 10
4. (a) Explain the architecture of iOS. 10
- (b) Explain various phases of development process of a mobile application. 10
5. (a) What is GPRS ? Explain its features. 10
- (b) What are adhoc networks ? Explain their advantages. 10